

Bank Marketing Data Audit and Exploratory Data Analysis (EDA)

1. Introduction

This report analyses the Bank Marketing dataset. It contains information of customer and campaign collected during a bank's direct marketing activities. The objective of this analysis is to assess data quality, explore the characteristics of the dataset, and identify meaningful patterns that can support decision-making in future.

2. Customer and Campaign Data

The dataset contains customer demographic information, financial details, and marketing campaign data collected by a banking institution during direct marketing activities. The dataset consists of 45,211 observations and 17 variables, including both numerical and categorical features.

Data exploration was conducted by examining data types, missing values, duplicate records, and descriptive statistics. The variable name "y" was renamed to "customer_response" which is a meaningful description of customer subscription behaviour. Since, there is no missing values or duplicate records were identified the imputation process is not required, and all observations were retained for analysis. These suggest that the dataset is suitable for exploratory data analysis.

3. Methodology

- **Data Loading and Inspection** – `pd.read_csv()`
- **Data Type Verification** – `dtypes`, `info()`, `head()`, `shape`, `info()`
- **Missing Value Assessment** – `isnull().sum()`
- **Duplicate Record Identification** – `duplicated().sum()`
- **Distribution Analysis** – `hist()`, `boxplot()`
- **Descriptive Statistical Analysis** – `describe()`, `mean()`, `median()`, `mode()`
- **Measures of Dispersion** – `range`, `var()`, `std()`
- **Distribution and Outlier Analysis** – `quantile()`, IQR method, `skew()`
- **Relationship Exploration** – `corr()`, Heatmap
- **Data Visualization** – Histogram, Box Plot, Correlation Heatmap

4.Visualizations

Box Plot- A box plot was used to identify outliers and find the spread of customer account balance and it was found balance variable contains a large number of outliers, indicating significant variation in customer financial positions and the presence of customers with exceptionally high balances. (Fig-1)

Histogram- A histogram was used to find the distribution of customer ages and identify the most common age groups, and it was found that customers are mostly in between 30yrs to 50yrs, with few customers are older in age, because of which we see slightly right-skewed distribution. (Fig-2)

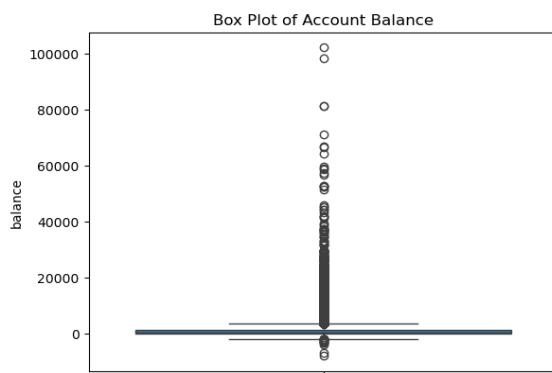


Fig-1

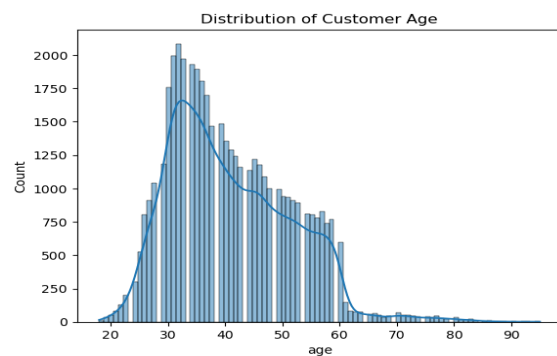


Fig-2

Correlation Heatmap- A correlation heatmap was used to find the relationships between numerical variables and identify any significant associations. It was found that the most numerical variables show weak correlations. However, a positive correlation (0.45) exists between pdays and previous, which shows a relationship between previous customer contacts and the number of days since the last contact. (Fig-3)

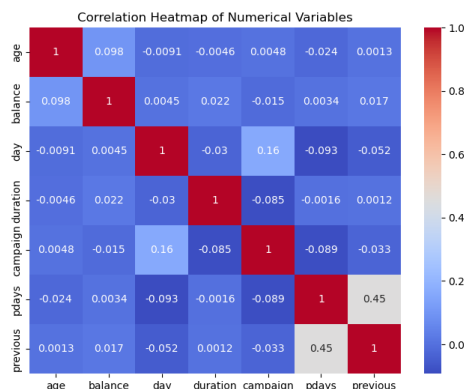


Fig-3

5. Key Insights and Recommendations

- The bank's customers are mainly middle-aged individuals, which shows that most of the customers of the bank are economically active.
- Customer account balances vary widely, which shows a diverse financial profile for targeted marketing.
- A small segment of customers holds high balances and can be targeted with premium banking products.
- Customer interaction history appears to influence future campaign outcomes, which shows the importance of maintaining relationship records.
- Marketing decisions should consider multiple customer factors rather than relying on a single variable.
- The bank can improve campaign effectiveness through better customer segmentation and personalized marketing strategies.

6. Conclusions

This exploratory analysis provided valuable insights of the Bank Marketing dataset, including customer demographics, account balances, and campaign interactions. The dataset was found to be complete and suitable for analysis. The findings demonstrate how customer data can support better marketing decisions and provide a foundation for further predictive analytics and customer segmentation studies.

AI Assistance Declaration: Generative AI tools were used to assist with report writing, code interpretation, and explanation of analytical findings. All results and interpretations were reviewed and validated by the student.